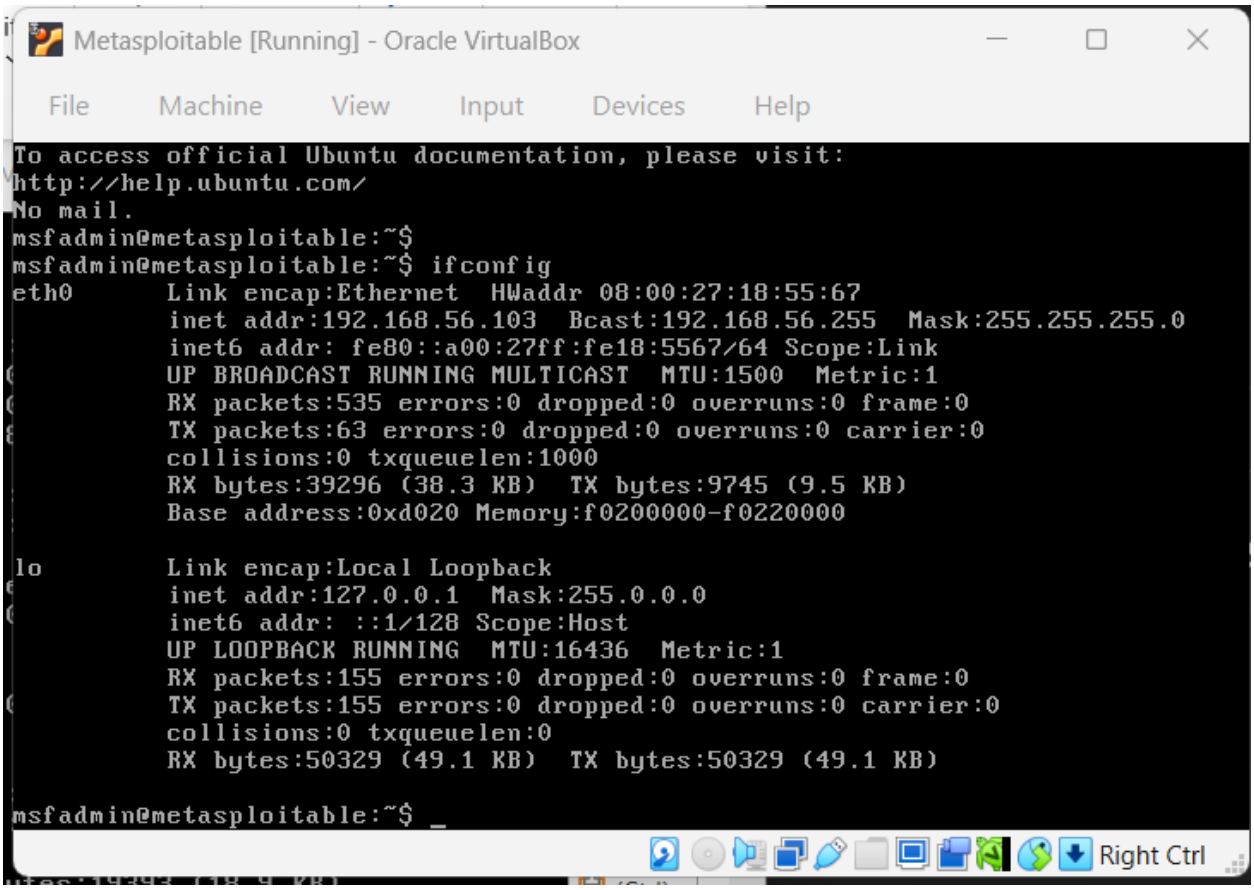
Nmap OS Detection.

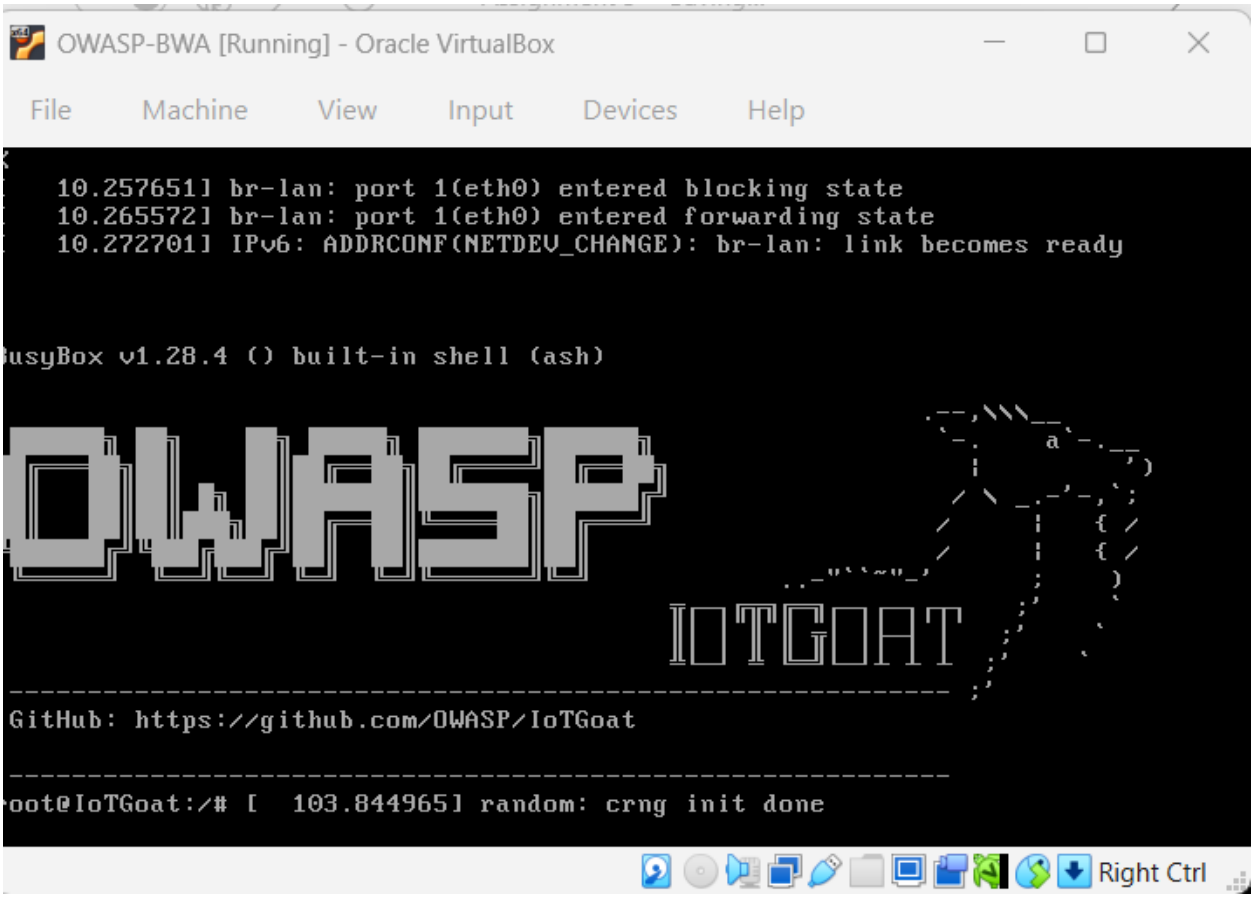
A ping sweep was conducted on the 192.168.56.0/24 network, identifying 5 active hosts including multiple virtual machines within the lab environment.

VM Machines ON:

- Metasploitable:



- OWASP


OWASP-BWA [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
10.257651] br-lan: port 1(eth0) entered blocking state
10.265572] br-lan: port 1(eth0) entered forwarding state
10.272701] IPv6: ADDRCONF(NETDEV_CHANGE): br-lan: link becomes ready
usyBox v1.28.4 () built-in shell (ash)
OWASP
IOTGOAT
GitHub: https://github.com/OWASP/IoTGoat
oot@IoTGoat:/# [103.844965] random: crng init done
Right Ctrl

Terminal showing IP on Kali:

```
root@kali: ~
Session  Actions  Edit  View  Help
┌──(root㉿kali)-[~]
└─# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.56.104  netmask 255.255.255.0  broadcast 192.168.56.255
        inet6 fe80::e29d:367f:ffe:8658  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:8a:35:d2  txqueuelen 1000  (Ethernet)
        RX packets 6  bytes 2176 (2.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 26  bytes 3818 (3.7 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(root㉿kali)-[~]
└─#
```

Discover Active Devices:

```
root@kali: ~
Session  Actions  Edit  View  Help
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 8  bytes 480 (480.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8  bytes 480 (480.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

┌──(root㉿kali)-[~]
└─# nmap -sn 192.168.56.104/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-04 20:45 -0400
Nmap scan report for 192.168.56.1
Host is up (0.00090s latency).
MAC Address: 0A:00:27:00:00:0C (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.0012s latency).
MAC Address: 08:00:27:3F:A3:40 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.102
Host is up (0.0024s latency).
MAC Address: 08:00:27:2F:74:52 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.103
Host is up (0.0023s latency).
MAC Address: 08:00:27:18:55:67 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.74 seconds

┌──(root㉿kali)-[~]
└─#
```

Run OS Detection to detect metasploitable:

```
┌──(root㉿kali)-[~]
└─# sudo nmap -sS 192.168.56.103 -O
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-04 20:51 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0018s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
```

```
root@kali: ~
Session  Actions  Edit  View  Help
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:18:55:67 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.o
rg/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.02 seconds

┌──(root㉿kali)-[~]
└─#
```

Run Aggressive Scan:

```
root@kali: ~
Session  Actions  Edit  View  Help

┌──(root㉿kali)-[~]
└─# sudo nmap -sV -A 192.168.56.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-04 20:54 -0400
Nmap scan report for 192.168.56.103
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.56.104
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
```

root@kali: ~

Session Actions Edit View Help

```
|_End of status
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
|_ssl-date: 2026-04-05T00:54:54+00:00; -1s from scanner time.
| sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|_    SSL2_RC2_128_CBC_WITH_MD5
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=O
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY,
 ETRN, STARTTLS, ENHANCEDSTATUSCODES, 8BITMIME, DSN
53/tcp   open  domain      ISC BIND 9.4.2
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-title: Metasploitable2 - Linux
```

root@kali: ~

Session Actions Edit View Help

```
|_http-title: Metasploitable2 - Linux
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
|   100003  2,3,4       2049/udp   nfs
|   100005  1,2,3      41704/tcp   mountd
|   100005  1,2,3      49236/udp   mountd
|   100021  1,3,4      43591/tcp   nlockmgr
|   100021  1,3,4      57388/udp   nlockmgr
|   100024  1          38092/udp   status
|_  100024  1          47964/tcp   status
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
```

root@kali: ~

Session Actions Edit View Help

```
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 8
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, SupportsTransactions, LongColumnFlag, S
upportsCompression, Speaks41ProtocolNew, SwitchToSSLAfterHandshake, ConnectWi
thDatabase
|   Status: Autocommit
|_  Salt: :k.CFs5KZc`;E(9$Eq+|
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-04-05T00:54:54+00:00; -1s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=O
COSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
```

```
root@kali: ~
Session  Actions  Edit  View  Help
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs
: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC:
 <unknown> (unknown)
| smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-04-04T20:54:45-04:00
|_smb2-time: Protocol negotiation failed (SMB2)
|_clock-skew: mean: 59m59s, deviation: 2h00m00s, median: -1s

TRACEROUTE
HOP RTT     ADDRESS
1   1.60 ms 192.168.56.103

OS and Service detection performed. Please report any incorrect results at ht
tps://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 24.46 seconds
```

In this lab, I successfully performed network discovery and operating system detection using Nmap within a virtual lab environment. By configuring the network correctly and conducting a ping sweep on the 192.168.56.0/24 range, I was able to identify multiple active hosts, including the target systems.

Using advanced Nmap scanning techniques such as SYN scan and aggressive scan, I gathered detailed information about the target machine, including its operating system, open ports, and running services. This process demonstrated the effectiveness of Nmap as a powerful tool for reconnaissance and system profiling in penetration testing.

Overall, this lab enhanced my understanding of network scanning, OS fingerprinting, and service enumeration, which are critical skills in cybersecurity for identifying potential vulnerabilities and assessing system security posture.